Retinal bottleneck in anatomically constrained CNNs

A reproduction of Lindsey et al. (ICLR 2019) on STL-10 and Imagenette

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Abstract

Lindsey et al. (2019) train a convolutional network on object recognition, split into a "retina" and a "ventral visual stream", and show that two classical facts of visual neuroscience emerge from a single model: the retinal output has center-surround receptive fields while the first cortical layer has oriented ones, provided that the retinal output is a narrow bottleneck; and the depth of the cortex determines how linear and information-preserving the retina is. I reproduce their experiments on STL-10 and Imagenette (grayscale, 96×96) with less data (4 500–8 500 training images against 45 000), fewer seeds (3 against 10; 5 in the bridge experiment) and a single laptop GPU (about 10 hours of training in total): 72 networks for the bottleneck sweep, 54 for a cortex-depth grid, 18 networks in the paper's own setting (CIFAR-10) as a positive control, and 72 for a bridge experiment that moves from that setting to mine one factor at a time. The accuracy claim (accuracy grows with bottleneck width and cortex depth) is reproduced. The retinal geometry is reproduced only partially: at the paper's headline configuration (one bottleneck channel, two cortical layers) the retinal orientation index is 1.47 on Imagenette (paper: 1.56) but 2.90 on STL-10, the CIFAR-10 control gives 1.60, and elsewhere in the sweep most retinal filters are oriented. Retinal linearity grows with cortex depth only at the tightest bottleneck, and the first cortical layer is more oriented than the retina at narrow bottlenecks. The bridge experiment is inconclusive on its registered measure; descriptively, the isotropic one-channel retina survives fewer data and a higher resolution, and the many dead channels follow the paper's head. I document each difference from the original setup and report what I could and could not isolate.

1 Introduction

Why do retinal ganglion cells respond to local contrast while neurons in the primary visual cortex (V1) respond to oriented edges? Classical theories of efficient coding explain each geometry separately, but not why two consecutive stages should differ. Lindsey et al. [1] propose that the difference follows from a *resource constraint*: the optic nerve carries the signal of roughly a hundred million photoreceptors through about one million fibres, so the retina is a dimensionality bottleneck. They model the visual system as a deep CNN trained for object recognition, vary the width of the retinal output and the depth of the cortex, and look at the receptive fields (RFs) that emerge. The assignment of this project is a *reproduction* of that paper on two datasets chosen by the instructor, STL-10 and Imagenette at 96×96, with the ablations of bottleneck width, retina depth and cortex depth, and three seeds per configuration. I call it *reduced-scale* only in two respects, which may affect the outcome: the amount of training data (4 500 and 8 500 images against 45 000) and the computing resources (3 seeds against 10, one laptop GPU, about 10 hours of training in total). The other differences, such as the resolution (higher than in the paper), the classifier head and the analysis, are not a matter of scale and are listed in Section 4. The aim is not to match the numbers of the paper but to measure carefully what survives the change of setting and to say why, or why I cannot say. The report is organised as follows: Section 2 gives the neuroscience needed to read the paper; Section 3 summarises its model and claims; Section 4 describes my setup and every difference from the original; Section 5 reports the results, including a positive control run in the original setting; Sections 6–8 discuss, list the limitations and conclude.

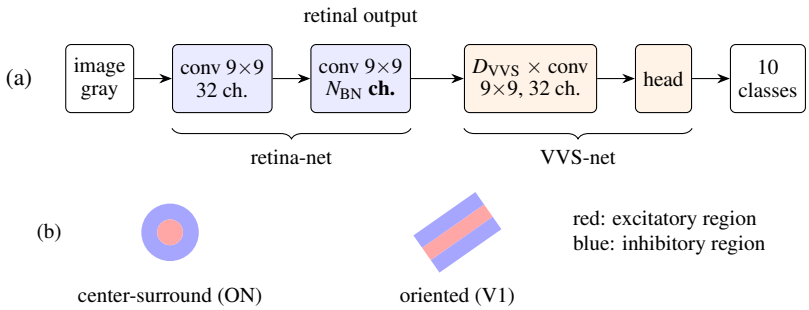


Figure 1: (a) The architecture of the paper and of this project: a retina-net whose output is a bottleneck of N_{BN} channels, followed by a cortex of D_{VVS} convolutions and a classifier head. (b) The two receptive-field geometries of the early visual system.

2 Background: the early visual system

From the eye to the cortex. Light is transduced by photoreceptors (rods and cones). Their signal passes through bipolar cells to the *retinal ganglion cells*, whose axons form the optic nerve. In humans there are of the order of 10^8 photoreceptors and 10^6 ganglion cells, a compression of about two orders of magnitude, so the optic nerve is a stringent bottleneck. The signal is relayed by the thalamus (lateral geniculate nucleus, LGN) to the primary visual cortex V1 and then along the *ventral stream* (V2, V4, inferotemporal cortex), which supports object recognition.

Receptive fields. The receptive field (RF) of a neuron is the part of the visual scene, and the kind of pattern, that drives it. Two geometries matter here (Fig. 1b).

- *Center-surround* RFs of retinal ganglion cells (and LGN): a small excitatory center surrounded by an inhibitory ring, or the opposite (ON and OFF cells). They respond to local contrast and not to uniform illumination, and act as a decorrelating (whitening) filter.
- *Oriented* RFs of V1 simple cells: elongated patches responding to an edge of a given orientation and position, well described by Gabor functions. Hubel and Wiesel [5] hypothesised that they are built by pooling center-surround inputs aligned along the edge; complex cells then pool simple cells and add position invariance.

The open question. Efficient-coding theories reproduce center-surround RFs (decorrelation of natural images under noise [6]) and oriented RFs (sparse coding [7]), but with different objectives and constraints for the two stages. Moreover, retinas differ across species: in primates the most common ganglion cell type is close to a linear pixel encoder (midget cells), while in the mouse the most common type is a nonlinear feature detector [8], and the frog retina detects behaviourally relevant stimuli. A single account should explain both the retina–V1 difference and the diversity across species.

3 The reference paper

Model. The paper trains a CNN on grayscale CIFAR-10 [2] (Fig. 1a). A *retina-net* of two convolutions (32 channels, then N_{BN} channels, the retinal output) feeds a *ventral-visual-system net* (VVS-net) of D_{VVS} convolutions with 32 channels followed by two dense layers (1024 and 10 units). All convolutions are 9×9, stride 1, with ReLU and no batch normalisation; training uses RMSProp for 20 epochs, batch size 32, Glorot-uniform initialisation [4], ten random initialisations per architecture (Keras/TensorFlow). The RF of a channel is estimated as the gradient of its activation at the image centre with respect to a blank image, equivalent to a spike-triggered average for a small perturbative stimulus.

Claims. (i) *A bottleneck at the retinal output* ($N_{\mathrm{BN}} = 4, 2, 1$) *yields center-surround retinal RFs and oriented RFs in the first cortical layer*; with $N_{\mathrm{BN}} = 32$ the retinal RFs are oriented. The RF geometry does not depend qualitatively on D_{VVS} (shown for $D_{\mathrm{VVS}} = 2$), except for $D_{\mathrm{VVS}} = 0$. (ii) *The cortex depth sets the linearity of the retina:* with a deeper cortex the retinal response is more linear and keeps more information about the image but separates the classes less; with a shallow cortex the retina is nonlinear and extracts features. (iii) *The accuracy increases with the bottleneck width and with the cortex depth* (their Fig. 1D).

Measures. The *orientation selectivity index* (OSI) is computed from the responses of an RF to rectangular bars of all widths, orientations and positions: the ratio between the strongest response at the preferred orientation and the strongest response at the orthogonal one. It is ≈ 1 for isotropic RFs and larger for oriented ones. The paper reports 1.56 ± 0.22 for the retinal output, 3.05 ± 0.30 for the first cortical layer, and 1.97 ± 0.08 for random-noise RFs. *Linearity* is the Pearson correlation between the true response of a retinal unit and its ridge-regression prediction from the image; *information* is the variance explained when the image is linearly reconstructed from the retinal output.

4 My reproduction: setup and differences

4.1 Hypotheses of the assignment

The assignment is phrased as three hypotheses: (H1) a narrow bottleneck with a deep cortex makes the retina learn center-surround filters; (H2) a wide bottleneck or a shallow cortex makes the retina learn oriented filters; (H3) the accuracy degrades little as the bottleneck narrows, until a minimum width. After reading the paper in full I note that H1 and H2 are stricter than claims (i) and (ii): the paper does not tie center-surround RFs to a deep cortex, nor oriented retinal RFs to a shallow one, and its cortex-depth effect is on the *linearity* of the retina. I therefore test both the hypotheses as written and the paper's own claims.

4.2 Data

STL-10 and Imagenette are decoded once (resize to 96, centre crop, grayscale) and kept as 8-bit tensors on the GPU. The test sets are the official STL-10 `test` split and the Imagenette `val` split. CIFAR-10 is not simply "easier": it has far more labels per class (5 000 against 500 for STL-10) but a much lower resolution. The two properties that matter for the mechanism studied here are the resolution, relative to the 9×9 kernels, and the amount of training data.

4.3 Architecture, training and differences from the paper

My network follows Fig. 1a with $N_{\mathrm{BN}} \in \{1, 2, 4, 8, 16, 32\}$, retina depth 1 or 2 (the last retina layer has N_{BN} channels) and $D_{\mathrm{VVS}} \in \{1, 2, 3, 4\}$. Table 2 lists every difference from the original setup.

	Resolution	Classes	Labelled train / test	Role here
CIFAR-10	32×32 colour	10	50 000 / 10 000	paper's data; positive control
STL-10	96×96 colour	10	5 000 / 8 000 (+100 000 unlabelled, unused)	main dataset
Imagenette	up to 160 px	10 easy ImageNet classes	9 469 / 3 925	main dataset

Table 1: Datasets. STL-10 [3] is a subset of ImageNet with few labels per class; Imagenette is a ten-class ImageNet subset by fast.ai. A part of the training set (10%) is held out for validation, so the networks are trained on about 4 500 (STL-10) and 8 500 (Imagenette) images, against 45 000 for CIFAR-10.

	Paper	This project
Data	grayscale CIFAR-10, 32×32, 50 000 images	grayscale STL-10 / Imagenette, 96×96, 4 500 / 8 500 images (assignment)
Cortex head	flatten, dense 1024, dense 10	2×2 max-pool after every cortical conv, global average pool, linear layer (a flatten + dense 1024 at 96×96 would have hundreds of millions of weights)
Cortex depth	$D_{\mathrm{VVS}} = 0$–4	1–4; $D_{\mathrm{VVS}} = 0$ not run (no meaningful head)
Optimiser	RMSProp, 20 epochs, batch 32, Glorot-uniform	the same (RMSProp $\rho = 0.9$, $\varepsilon = 10^{-7}$, lr 10^{-3}); PyTorch
Not stated in the paper	input scaling, augmentation, checkpoint	input centred as $(x - 0.5)/0.25$ inside the model; random horizontal flip; checkpoint with best validation accuracy
Collapsed runs	not discussed	a run whose best validation accuracy is below 0.2 is retrained once with lr/3; both kept
RF probe	gradient at a blank image	gradient averaged over 256 natural images (a flat image leaves 77% of my retinal channels silent)
Seeds and compute	10 initialisations per architecture; hardware not stated	3 seeds (5 in the bridge experiment); one laptop GPU (RTX PRO 2000, 8 GB), about 10 h of training in total
Extra analyses	separability, untied retina, LGN, simple/complex cells, normalisation	not reproduced

Table 2: Differences between the original setup and mine.

4.4 Receptive-field analysis

The RF of each retinal channel is the gradient of its activation at the image centre with respect to the pixels, averaged over 256 natural images; the input is centred exactly as in the forward pass of the network. The OSI uses the paper's definition. In addition, every RF is fitted with an isotropic difference of Gaussians (DoG) and with a Gabor function, and labelled *center-surround* (OSI < 2 and a DoG fit with $R^2 \geq 0.5$ not worse than the Gabor by more than 0.05), *oriented* (OSI ≥ 2, Gabor $R^2 \geq 0.5$ and better than the DoG), *dead* (zero gradient) or *other*. The thresholds were fixed on synthetic filters before looking at the learned ones (the classifier recovers the true class of 40/40 synthetic filters per class under 10% noise). Because the labels depend on thresholds and fits, I also report the OSI itself, which is threshold free.

4.5 Experiments

- **Sweep (72 networks).** Two datasets × 12 configurations × 3 seeds: $N_{\mathrm{BN}} \in \{1, \ldots, 32\}$ at retina depth 2 and cortex depth 2, plus $N_{\mathrm{BN}} \in \{2, 16\}$ with retina depth 1 or cortex depth 4. 3 of 72 runs collapsed and were retrained.
- **Cortex-depth grid (54 networks).** $N_{\mathrm{BN}} \in \{1, 4, 32\}$ at $D_{\mathrm{VVS}} \in \{1, 3, 4\}$ (depth 2 comes from the sweep); 9 runs collapsed at the first attempt, 7 of them at $N_{\mathrm{BN}} = 1$.
- **Linearity, information and first cortical layer** on all 126 networks, without new training.
- **Positive control (18 networks).** The architecture of the paper, with flatten and dense layers and no pooling, on grayscale CIFAR-10 at 32×32 with my training recipe: $N_{\mathrm{BN}} \in \{1, 4, 32\}$, $D_{\mathrm{VVS}} \in \{2, 4\}$, 3 seeds. The prediction and its decision rule were written before the runs.
- **Recipe check (4 networks).** CIFAR-10, $N_{\mathrm{BN}} \in \{4, 32\}$, seed 0, to see whether my unstated choices (input centring, flip, checkpoint) matter.
- **Bridge experiment (72 networks).** Starting from the control, one factor at a time towards my setup: amount of data, image content, classifier head, resolution (Section 5.7). $N_{\mathrm{BN}} \in \{1, 4, 32\}$,

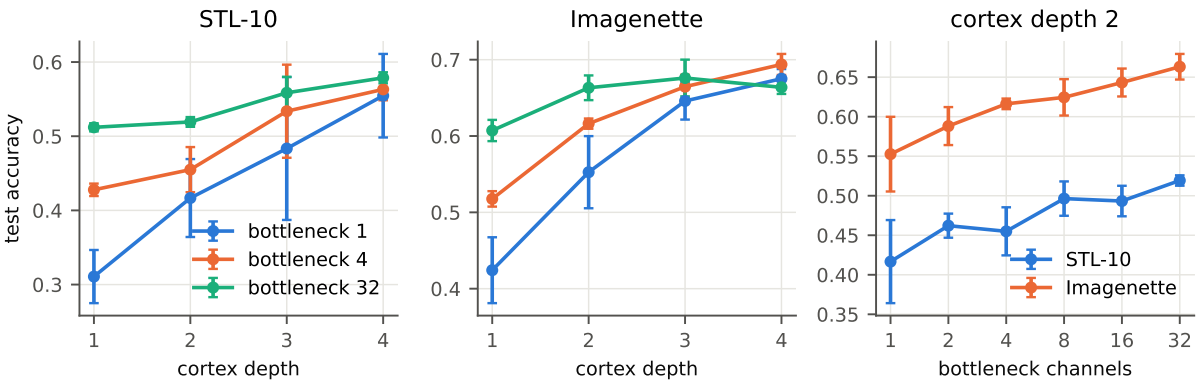


Figure 2: Test accuracy (mean ± s.d. over 3 seeds) against cortex depth for three bottlenecks (left, centre) and against bottleneck width at cortex depth 2 (right).

Dataset	N_{BN}	depth 1	depth 2	depth 3	depth 4
STL-10	1	0.311 ± 0.036	0.417 ± 0.053	0.483 ± 0.096	0.555 ± 0.056
STL-10	4	0.428 ± 0.008	0.455 ± 0.030	0.534 ± 0.063	0.563 ± 0.015
STL-10	32	0.512 ± 0.005	0.519 ± 0.007	0.559 ± 0.022	0.579 ± 0.007
Imagenette	1	0.424 ± 0.043	0.553 ± 0.047	0.646 ± 0.024	0.675 ± 0.013
Imagenette	4	0.518 ± 0.010	0.616 ± 0.007	0.665 ± 0.016	0.694 ± 0.014
Imagenette	32	0.607 ± 0.014	0.663 ± 0.016	0.676 ± 0.024	0.664 ± 0.009

Table 3: Test accuracy (mean ± s.d. over 3 seeds) by cortex depth. Part of the deep $N_{\mathrm{BN}} = 1$ runs are reruns with lr/3 after a collapse.

cortex depth 2, 5 seeds; the two endpoints reuse their first 3 seeds. The design, the measure and the decision rule were written before the runs.

5 Results

5.1 Accuracy (claim iii)

Test accuracy increases with the cortex depth in all six dataset–bottleneck cells (Fig. 2, Table 3): at $N_{\mathrm{BN}} = 1$ from 0.31 to 0.56 on STL-10 and from 0.42 to 0.68 on Imagenette between depth 1 and 4. At depth 2 it increases with the width, from 0.417 to 0.519 (STL-10) and from 0.553 to 0.663 (Imagenette) between 1 and 32 channels, with small non-monotonic steps. I do not observe a knee: going from 32 to 1 channel costs about ten points on both datasets, and no narrower bottleneck comes within one point of the widest one. Both parts of claim (iii) are reproduced; H3 as formulated (little loss until a minimum width) is not. Absolute accuracies are low (0.31–0.69) because of the small training sets.

5.2 Retinal receptive fields (claim i)

Geometry. Of the 702 retinal channels of the sweep, 22 (3.1%) are labelled center-surround, 270 oriented, 316 other and 94 dead (13%) (Fig. 3; Table 4 lists the 378 channels of retina depth 2 and cortex depth 2). Even the unlabelled quantity points the same way: the median OSI of the 608 live channels is 2.5, against 1.56 in the paper, and 37% have OSI < 2. The label is conservative (the 22 center-surround channels have DoG and Gabor R^2 within 0.10 of each other, and isotropic filters with a poor DoG fit are labelled "other", median OSI 1.8), so I rely on the OSI as well. Fig. 5 shows examples.

Dataset	N_{BN}	Channels	C-S	Orient.	Other	Dead	Median OSI	OSI<2
STL-10	1	3	0	2	1	0	3.23	1/3
STL-10	2	6	1	5	0	0	2.33	1/6
STL-10	4	12	0	6	5	1	2.71	5/11
STL-10	8	24	0	15	4	5	3.26	2/19
STL-10	16	48	1	22	15	10	2.63	14/38
STL-10	32	96	2	49	22	23	2.85	23/73
Imagenette	1	3	1	0	2	0	1.51	3/3
Imagenette	2	6	3	3	0	0	2.33	3/6
Imagenette	4	12	0	10	1	1	2.87	0/11
Imagenette	8	24	1	13	7	3	3.51	6/21
Imagenette	16	48	2	28	13	5	2.86	14/43
Imagenette	32	96	2	54	26	14	3.56	18/82

Table 4: Labels of the retinal channels of the sweep (retina depth 2, cortex depth 2; C-S = center-surround), median OSI of the live channels and number of live channels with OSI < 2.

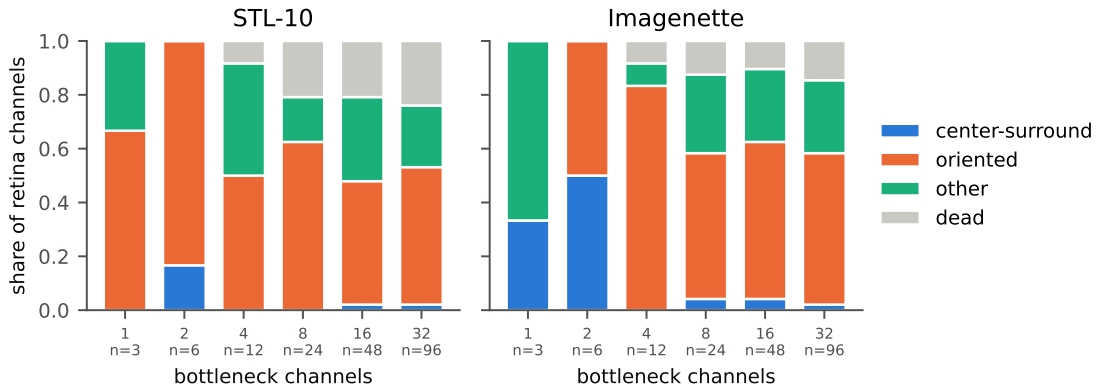


Figure 3: Composition of the retinal channels by label for each bottleneck (retina depth 2, cortex depth 2). n is the number of retinal channels (3 seeds $\times$ bottleneck width).

The paper's headline configuration ($N_{\mathrm{BN}} = 1$, cortex depth 2) gives a live retinal channel per network. On Imagenette their OSI is 1.22, 1.51 and 1.69 (mean 1.47, close to the 1.56 of the paper; all three below 2) but on STL-10 it is 1.84, 3.23 and 3.62 (mean 2.90). Pooling the narrow bottlenecks (1–2 channels, cortex depth 2), 6 of 9 live channels have OSI < 2 on Imagenette against 32 of 125 at 16–32 channels; on STL-10 the figures are 2 of 9 against 37 of 111, i.e. no effect (Fig. 4). With at most 9 channels at the narrow end, the effect is therefore dataset dependent and suggestive only.

H1 and H2 as written. In the configuration of H1 ($N_{\mathrm{BN}} = 2$, cortex depth 4) only 1 of 6 channels is center-surround on each dataset, 3 of 6 are oriented and the median OSI is 2.7 (STL-10) and 2.0 (Imagenette): H1 is not reproduced. For H2, bottleneck 32 has 49/96 and 54/96 oriented channels, but narrow bottlenecks are not less oriented on STL-10 (7 of 9 channels at 1–2 channels) and less so on Imagenette (3 of 9). With the bottleneck fixed, cortex depth 2 has more oriented channels than cortex depth 4 in three of four comparisons and a tie in the fourth (e.g. $N_{\mathrm{BN}} = 16$: 22/48 against 14/48 on STL-10, 28/48 against 16/48 on Imagenette), consistent with H2 as worded; this is not a claim of the paper, which reports no dependence of the geometry on the depth. With retina depth 1 most filters are "other" and the 9×9 RF is too small for the fits to be reliable.

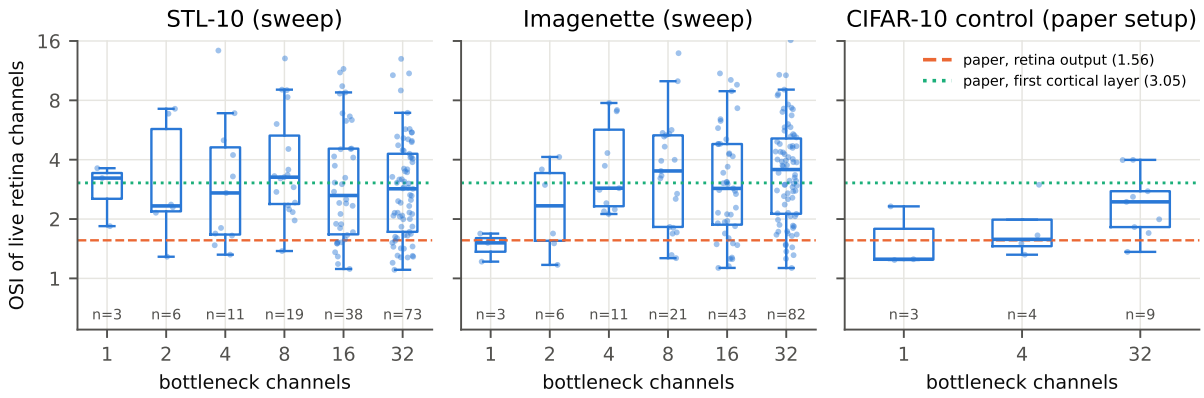


Figure 4: OSI of the live retinal channels at cortex depth 2: sweep on STL-10 and Imagenette, and CIFAR-10 control (bottleneck widths 1, 4, 32). Dashed: paper's retinal OSI; dotted: paper's first cortical layer.

5.3 First cortical layer (claim i, second half)

The first cortical layer has all its 32 channels alive and median OSI between 2.4 and 5.3, around the 3.05 of the paper (Fig. 6). It is more oriented than the retinal output at narrow bottlenecks (1–4 channels) in 4 of 6 cells ($N_{\mathrm{BN}} = 1$: 3.3 against 1.5 on Imagenette, 5.3 against 2.9 on STL-10), while at wide bottlenecks the retina is itself oriented and there is no difference (4 of 12 cells in total). The retina-to-cortex step from isotropic to oriented RFs is therefore visible at the tightest bottleneck, in agreement with the paper.

5.4 Linearity and information (claim ii)

Linearity of the retinal response (Fig. 7, top) *decreases with the bottleneck width* at depth 2 (Imagenette 0.73 at 1 channel to 0.42 at 32; STL-10 0.62 to 0.39), as in the paper. It *increases with the cortex depth only at* $N_{\mathrm{BN}} = 1$: from 0.52 to 0.76 (STL-10) and from 0.65 to 0.76 (Imagenette) between depth 1 and 4, which is the case the paper highlights (about 0.9 of the variance explained at one channel with the deepest cortex). At $N_{\mathrm{BN}} = 4$ and 32 the trend is flat or non-monotonic with a maximum at depth 2 (the paper reports a maximum at depth 1 without bottleneck). A deeper cortex also leaves fewer live retinal channels at width 32 (STL-10: 27 at depth 1, 12.7 at depth 4), so these means are over a shrinking set. The information kept by the retina (local reconstruction of the centre pixel from a 9×9 window of retinal output; Fig. 7, bottom) increases with depth in 4 of 6 cells (e.g. STL-10 at one channel from 0.08 to 0.36) and is flat in the other two; it grows with the width, as it must. Claim (ii) is reproduced in the tightest-bottleneck case, which is also the one where the retina has a single channel and where the paper finds the strongest effect.

5.5 Positive control in the paper's own setting

To separate the effect of my setup from the effect of my analysis, I trained the architecture of the paper on grayscale CIFAR-10 (Table 5). The recipe is fragile in my hands: accuracy is 0.56–0.67 (similar to the range of 0.5–0.7 on the axis of their Fig. 1D), 6 of 18 runs collapsed at the first attempt (5 at one channel), and 67% of the retinal channels are dead (149 of 222) against 13% in the sweep.

Headline configuration. At one bottleneck channel and cortex depth 2, the three seeds give OSI 1.24, 1.25 and 2.32 (mean 1.60; 1.61 with a flat-image probe), against 1.56 ± 0.22 in the paper. The control reproduces the number of the paper in its own setting.

Pre-registered prediction. Before running I predicted that the pipeline would find at least 30% center-surround channels among the live ones at bottleneck ≤ 4 with cortex depth 4, against less than 10% at

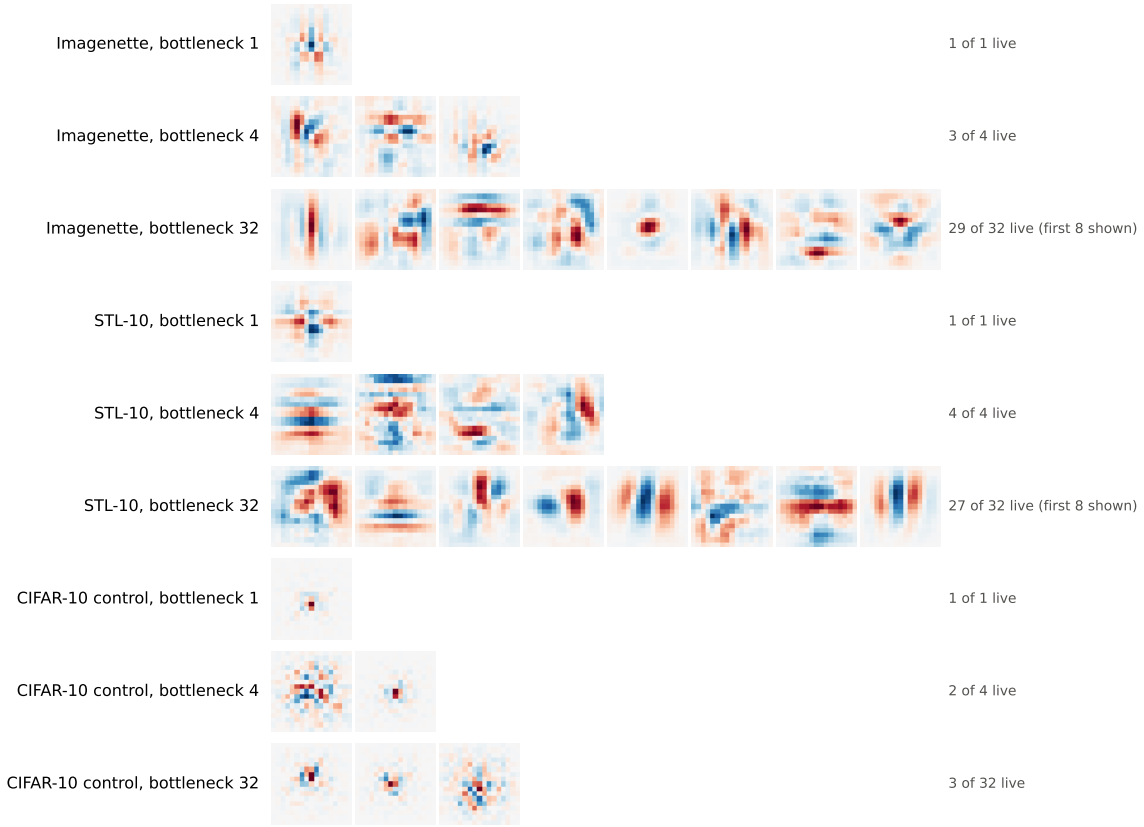


Figure 5: Retinal receptive fields (seed 0, cortex depth 2; red positive, blue negative, each scaled to its own maximum). Top: Imagenette and STL-10 sweep; bottom: CIFAR-10 control. At the narrowest bottleneck the single channel is compact and isotropic on Imagenette and on CIFAR-10; with 32 channels most filters are elongated.

bottleneck 32 with cortex depth 2. The outcome is 2 of 7 (29%) and 1 of 9 (11%): both bounds are missed, narrowly, and on a handful of channels; the OSI-based fallback also fails (3 of 7 against 4 of 9). By the rule as written the control does not license attributing the missing effect in the sweep to the data or the architecture. The prediction was written for cortex depth 4, following my H1, whereas the paper shows its results at depth 2. Looking at depth 2 afterwards (a choice made after seeing the data), 5 of 7 live channels at $N_{\mathrm{BN}} = 1, 4$ have OSI < 2 (median 1.50) against 4 of 9 at 32 channels (median 2.44), the direction of the paper; at depth 4 there is no contrast.

5.6 Recipe check

I tested whether the unstated choices of my recipe explain the many dead channels (CIFAR-10, $N_{\mathrm{BN}} = 32$, depth 2, seed 0): with my recipe 28 of 32 retinal channels are dead; without the input centring 27; without centring and flip 26–27; and the final epoch rather than the best-validation checkpoint gives the same count. At $N_{\mathrm{BN}} = 4$ without centring and flip, 2 of 4 channels are dead. None of these choices matters, so the dead channels are a property of the paper's recipe in my implementation as far as I can tell (one seed, four runs). The validation curves are noisy (peaks of 0.55–0.58 followed by oscillations), typical of RMSProp at lr 10^{-3}.

5.7 Bridge experiment: which difference matters?

The control cannot say which of the differences between the paper's setup and mine removes the effect, because data, resolution and head change together. I therefore walked from the control to my setup one factor at a time (Table 6): fewer images (`data`), STL-10 images instead of CIFAR-10 (`stl32`), my pooling

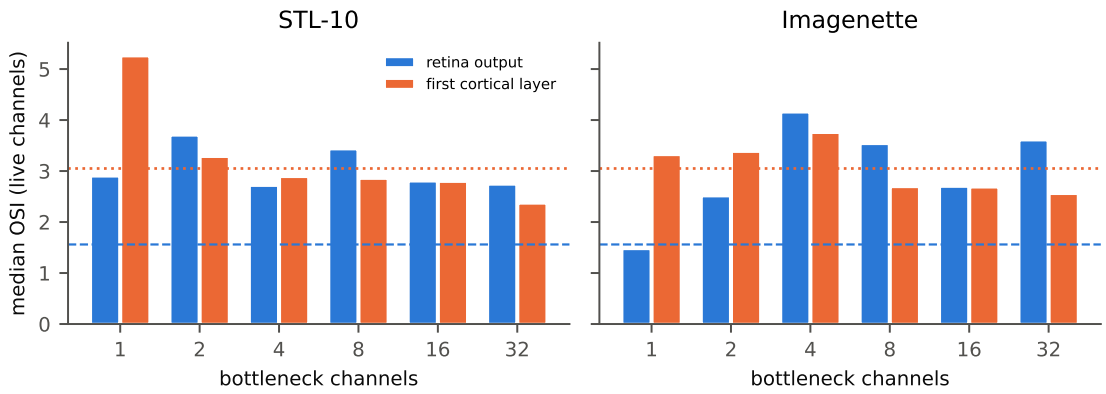


Figure 6: Median OSI of the live channels of the retinal output and of the first cortical layer (cortex depth 2). Dashed: paper's retina (1.56); dotted: paper's first cortical layer (3.05).

N_{BN}	Depth	Test acc.	Collapsed	Live/ch.	C-S	Median OSI	OSI<2
1	2	0.672 ± 0.004	3	3/3	1	1.25	2/3
1	4	0.652 ± 0.079	2	3/3	0	2.31	1/3
4	2	0.605 ± 0.024	0	4/12	2	1.58	3/4
4	4	0.555 ± 0.022	0	4/12	2	1.85	2/4
32	2	0.594 ± 0.021	0	9/96	1	2.44	4/9
32	4	0.573 ± 0.092	1	50/96	1	2.55	18/50

Table 5: CIFAR-10 control (paper's architecture, natural-image probe): collapsed first attempts, live channels out of the channels of the 3 seeds, center-surround labels, and OSI.

head (`stl32_pool`) and 96 pixels (`sweep`), plus a side branch with my head on CIFAR-10 (`head`). The paper's head cannot be run at 96×96 (about 300 million weights), so the resolution is tested only with the pooling head. Before running I fixed the measure, Δ = median OSI of the live channels at $N_{\mathrm{BN}} = 32$ minus that at $N_{\mathrm{BN}} \in \{1,4\}$ (positive when the narrow retina is more isotropic), and the rule: the effect is present if $\Delta \geq 0.5$ and its 90% interval over seeds lies above 0, and the factor to blame is the first step at which it goes from present to absent, provided that the control itself shows it.

Registered rule. The control does not show the effect ($\Delta = 0.24$, interval -0.66 to 1.20): at $N_{\mathrm{BN}} = 32$ only 15 of its 160 channels are alive, so the wide retina has nothing to contrast with. By the rule the experiment is therefore inconclusive, and no factor can be blamed on this measure; the one step formally "present" (`stl32`, $\Delta = 0.52$) is not interpretable.

Post hoc. Looking at the one-channel retina directly (chosen after seeing the table, 5 networks per condition), it is isotropic (OSI < 2) in 4 of 5 networks of the control, 3 of 5 with fewer data, 4 of 5 with STL-10 images and the paper's head, and 4 of 5 with my head on CIFAR-10, but oriented in 4 of 5 (`stl32_pool`) and 3 of 5 (`sweep`) with STL-10 images and my head: it turns oriented only when these two are combined (15 of 20 isotropic networks elsewhere against 3 of 10, Fisher exact $p = 0.045$, uncorrected). The dead channels follow the head and not the data: 66–91% at $N_{\mathrm{BN}} = 32$ with flatten and dense layers, 9–29% with pooling and global average pooling. The STL-10 headline value of Section 5 (2.90) is for seeds 0–2; the two added seeds give 2.95 and 1.21.

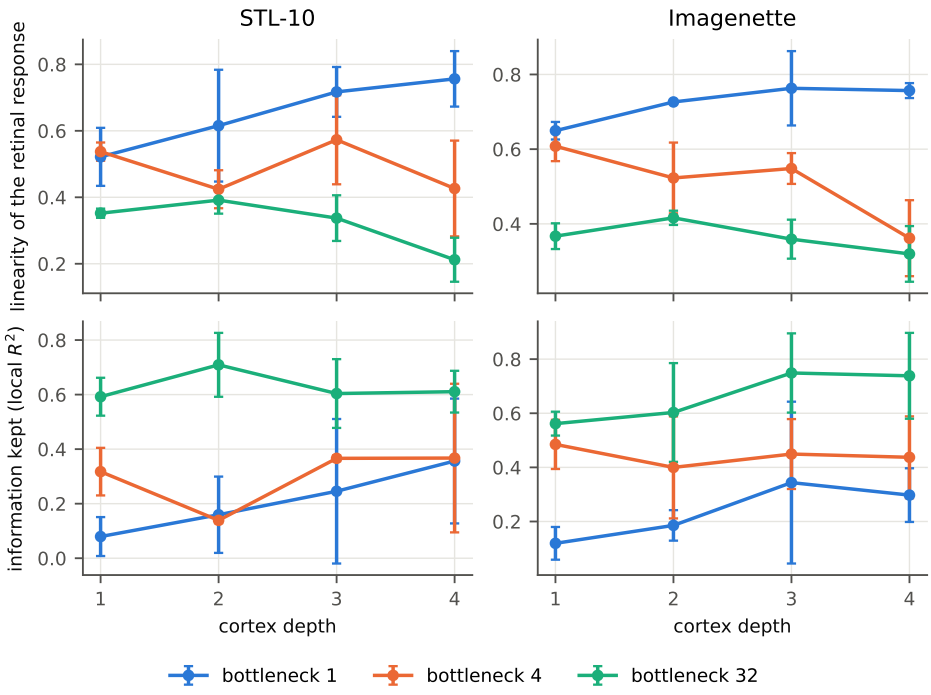


Figure 7: Linearity of the retinal response and local information kept by the retina against cortex depth (mean ± s.d. over 3 seeds, live channels only).

Condition	Images	Train	Res.	Head	Dead, N_{BN}=32	OSI, N_{BN}=1 (<2)	Δ [90% interval]
control	CIFAR-10	45 000	32	flatten + FC	145/160	1.25 (4/5)	0.24 [−0.66, 1.20]
data	CIFAR-10	4 500	32	flatten + FC	124/160	1.48 (3/5)	−0.05 [−0.58, 0.63]
stl32	STL-10	4 500	32	flatten + FC	106/160	1.41 (4/5)	0.52 [0.24, 0.65]
stl32_pool	STL-10	4 500	32	pool + GAP	26/160	2.69 (1/5)	−0.05 [−0.37, 0.31]
sweep	STL-10	4 500	96	pool + GAP	46/160	2.95 (2/5)	0.27 [−0.33, 0.75]
head	CIFAR-10	45 000	32	pool + GAP	14/160	1.36 (4/5)	0.44 [−0.13, 0.73]

Table 6: Bridge experiment (retina depth 2, cortex depth 2, 5 seeds): channels dead out of 160 at $N_{\text{BN}} = 32$, median OSI of the live channel of the 5 one-channel networks (in brackets: how many have OSI < 2), and the contrast Δ of the rule with its 90% interval.

6 Discussion

What reproduces. (iii) accuracy increases with the width and the depth; (ii) linearity decreases with the width and increases with the depth at the tightest bottleneck; (i) the first cortical layer is more oriented than the retina at narrow bottlenecks, with OSI values around those of the paper; and, in the original setting, the OSI of the one-channel retina (1.60 against 1.56). Table 7 summarises the status of each claim.

What does not. In my setup the retinal filters are mostly oriented, the center-surround labels are rare, and the one-channel retina is isotropic on Imagenette (1.47) but not on STL-10 (2.90). The linearity effect of the depth does not appear at 4 and 32 channels. The hypotheses H1 and H2, as written, are not confirmed, although they were not exactly the claims of the paper.

Why. The bridge experiment (Section 5.7) narrows the question without closing it. Its registered measure is inconclusive, because the wide retina of the control is 91% dead, but the one-channel retina, looked at directly and after the fact, rules out three candidates: it is not made oriented by the smaller amount of data, by the higher resolution (with the pooling head) or by my head alone. It turns oriented only when

Claim of the paper	Status	Evidence in this project
(i-a) Narrow bottleneck: isotropic (center-surround) retinal RFs	Partly	one-channel retina OSI 1.47 (Imagenette), 2.90 (STL-10), 1.60 (CIFAR-10 control); paper 1.56; sweep mostly oriented
(i-b) Oriented RFs in the first cortical layer	Reproduced	more oriented than the retina at narrow bottlenecks in 4 of 6 cells; OSI 2.4–5.3 (paper 3.05)
(ii-a) Retinal linearity decreases with the width	Reproduced	Imagenette 0.73 $\rightarrow$ 0.42, STL-10 0.62 $\rightarrow$ 0.39 (1 to 32 channels)
(ii-b) Linearity increases with the cortex depth	Partly	only at one channel (0.52 $\rightarrow$ 0.76, 0.65 $\rightarrow$ 0.76); flat or non-monotonic at 4 and 32
(ii-c) The retina keeps more information with a deeper cortex	Partly	4 of 6 cells (local proxy)
(iii-a) Accuracy increases with the cortex depth	Reproduced	6 of 6 cells, e.g. 0.31 $\rightarrow$ 0.56 (STL-10, one channel)
(iii-b) Accuracy increases with the width	Reproduced	0.417 $\rightarrow$ 0.519 (STL-10), 0.553 $\rightarrow$ 0.663 (Imagenette); no knee

Table 7: Summary of the claims of the paper against my results (my setup; the CIFAR-10 control is quoted where relevant).

STL-10 images are combined with the pooling head, which points to an interaction between image content and head that I did not test; with five networks per condition and a comparison chosen after the fact ($p = 0.045$, uncorrected) I treat it as a lead and not as an answer. My earlier guess about the resolution (from a 32×32 pilot run with an earlier, wrong RF probe) is not supported.

Dead channels. Retinal channels that never fire are common (13% in the sweep, 67% in the control). The bridge experiment shows that they follow the classifier head (66–91% dead at $N_{\mathrm{BN}} = 32$ with the paper's flatten and dense layers, 9–29% with pooling and global average pooling). A 32-channel retina that is mostly dead behaves like a narrow one, which blurs the comparison between widths and is why the control cannot show a wide-versus-narrow contrast; the paper does not discuss this.

Relation to the assignment. H1: not reproduced as written (1 of 6 center-surround channels in the configuration of the hypothesis), with the isotropic one-channel retina on Imagenette and in the control as the exception. H2: partly supported as worded (fewer oriented channels with the deeper cortex in 3 of 4 comparisons), but the dependence on the bottleneck width is absent on STL-10 and the claim is not the paper's. H3: not confirmed as a threshold; accuracy decreases gradually with the width.

7 Limitations

- Three seeds instead of ten (five in the bridge experiment); few live channels at the narrow bottlenecks (one per network at $N_{\mathrm{BN}} = 1$), so every statement about them is suggestive.
- Retinal filters are labelled with thresholds set on synthetic filters and not validated on learned ones; 77% of the channels are silent on a flat image, so the paper's blank-image probe cannot be used and the natural-image probe differs from it.
- The head differs from the paper's, the depth $D_{\mathrm{VVS}} = 0$ is not run, and the information measure is a local proxy of the paper's whole-image reconstruction; class separability, the untied retina, the LGN network, simple/complex cells and local normalisation are not reproduced.
- 9 of the 54 depth-grid networks and 3 of the 72 sweep networks collapsed and were retrained with lr/3, which mixes two learning rates in the deep one-channel cells.

- The recipe check is a single seed. The bridge experiment is inconclusive on its registered measure; its one-channel pattern rests on five networks per condition and a comparison chosen after the fact, and a path may hide interactions.

8 Conclusions

In my setting (two datasets of 96×96 images with 4 500–8 500 training examples, a pooling head, three seeds) I reproduce the accuracy claim of Lindsey et al., the dependence of retinal linearity on the bottleneck width, the increase of linearity with the cortex depth at the tightest bottleneck, and the step from isotropic to oriented RFs between the retina and the first cortical layer. I reproduce the center-surround retina only partially: at the paper's headline configuration the one-channel retina has an OSI of 1.47 on Imagenette (1.56 in the paper) and 1.60 in the CIFAR-10 control, but 2.90 on STL-10, and elsewhere in the sweep retinal filters are mostly oriented. A positive control in the original setting shows that my analysis can return the paper's number. A bridge experiment from that setting to mine is inconclusive on its registered measure; descriptively, neither fewer data nor a higher resolution makes the one-channel retina oriented, and the many dead channels come from the paper's head. The main contribution of the project is therefore a map of what survives the change of setting and what does not, with each difference from the original setup documented.

A Reproducibility and corrections

Hardware and time: one laptop GPU (NVIDIA RTX PRO 2000 Blackwell, 8 GB), about 10 hours of training in total (sweep 2.5 h, depth grid 2.1 h, CIFAR-10 control 3.1 h, recipe check 0.5 h, bridge experiment about 2 h). All code is in `retina_bottleneck.ipynb`; results are in `results/` (`results.csv`, `reruns.csv`, `filters.csv`, `depth_grid.csv`, `control_cifar10.csv`, `control_filters.csv`, `linearity_info_layer1.csv`, `recipe_check.jsonl`, `bridge/`, `bridge_filters.csv`); figures and tables of this report are generated by `report/make_figures.py`. A first analysis probed the retina with raw pixels, without the input centring used in the forward pass of the networks. The error was found while re-reading the paper and corrected; the counts changed (center-surround channels 11 → 22 of 702, dead 137 → 94) but not the picture, and the old outputs are kept in `results/*_rawprobe.csv`. All numbers in this report use the corrected probe.

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